

SANGAM KUMAR VERMA

Ramgarh, Jharkhand, India

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CAREER OBJECTIVE

- Final-year Electronics and Communication Engineering student skilled in **embedded systems, sensors, IoT, C, and Python**. Strong interest in **embedded development, sensor-based systems, debugging, and hardware-software integration**, seeking opportunities to apply practical skills in an industry environment.

ACADEMIC QUALIFICATIONS

- Bachelor of Technology – Electronics and Communication Engineering** Sep 2023 - April 2027
NIST University, Berhampur | CGPA: **9.81**
- Intermediate (Class XII)** 2022
DAV Public School, Ramgarh | Percentage: **86.7%**
- Matriculation (Class X)** 2020
DAV Public School, Ramgarh | Percentage: **90.4%**

TECHNICAL SKILLS

- Programming** : C, Python, Verilog HDL, MATLAB
- Semiconductor / EDA** : Cadence Virtuoso, Xilinx Vivado, CST Studio Suite
- Hardware & Embedded** : STM32N657, ESP32, Arduino, Sensors, GSM/GPS Interfaces
- Analysis** : Digital Circuits, Circuit Simulation, Signal Analysis, Debugging
- Development Tools** : Arduino IDE, Visual Studio Code, Tinkercad, SNAP

RELEVANT EXPERIENCE

- Summer Research Internship – NIT Jamshedpur** May-June 2026
 - Worked on skin Cancer Detection Using Terahertz Metamaterial Absorber
- Summer Research Internship – NIT Jamshedpur** May-June 2025
 - Worked on Cadence Virtuoso-based analog and mixed-signal circuit design and simulation.
- Agniirva Space Internship Program – IN-SPACe** 2024

PROJECT EXPERIENCE

- Skin Cancer Detection Using Terahertz Metamaterial Absorber** May 2026
 - Designed and simulated a terahertz metamaterial absorber using CST Studio Suite for biomedical sensing.
 - Evaluated absorber performance through electromagnetic analysis and optimized sensing characteristics.
 - Applied machine learning techniques to classify skin tissue using terahertz biosensor data.
- SAR-based Change Detection using Log-Ratio Analysis on Multi-Temporal Sentinel-1 Data** Jan 2026
 - Performed change detection on multi-temporal Sentinel-1 SAR imagery using log-ratio analysis.
 - Applied calibration, speckle filtering, and co-registration techniques for accurate analysis.
 - Analyzed land surface changes for disaster monitoring and environmental applications.
- Automatic Railway Gate System using SMS** Mar 2025
 - Designed an automated railway gate control system using ESP32, servo motor, and GSM module.
 - Improved railway crossing safety through SMS-based notification and automation.
- Automatic Accident Detection System using a Black Box** Dec 2024
 - Developed an embedded safety system using ESP32, GPS, GSM, vibration sensor, MQ2, and DHT11.
 - Implemented real-time accident detection with automatic emergency alert transmission.

CERTIFICATIONS & TRAINING

- PwC Advisory Launchpad Program, 2026** – PwC AC(India) Feb-Aug 2026
- Faculty Development Program (FCSD-2026), NIT Patna & NIT Jamshedpur** May-Jun 2026
 - Completed a two-week FDP on FPGA-Based Communication System Design.
 - Covered FPGA Architecture, Verilog HDL, VLSI, Xilinx Tools, AI/ML on FPGA, and Embedded Systems.
- MOOC on Earth Observation for Climate Action** – IIRS Dehradun 2023

ACHIEVEMENTS

- Selected for the **TRON Programming Contest 2026, Japan**, organized by the TRON Forum and STMicroelectronics; shortlisted after the initial screening for the development phase. 2026
- Winner – Inter-College Technical Paper Presentation, RIT Berhampur 2023
- Winner – Intra-College Hardware Model Presentation, REC Club (NIST) 2023

VOLUNTEER EXPERIENCE

- President, Renewable Energy Club, NIST** – Led initiatives focused on **renewable energy and embedded systems**, coordinating technical projects and student activities. 2025–2026
- Organized and hosted an **IoT Workshop and Technical Event** for **150+ students**, conducting hands-on sessions and introducing participants to IoT, embedded systems, and emerging technologies. 2026